

Functional Features Overview

Date: 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

The Emotion Detection and Learning Support Engine provides a comprehensive set of functional features that enable emotion detection from user text, personalized learning guidance, prediction logging, and analytics visualization. All planned functional features have been successfully implemented and verified.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Text Input	Accepts textual input describing user's emotions.	Streamlit UI	Done	High
2	Emotion Detection	Predicts emotion using trained BiLSTM model.	Emotion Detection Engine	Done	High
3	NLP Text Preprocessing	Converts user text into token sequences.	Preprocessing Module	Done	Medium
4	Confidence Score	Displays prediction confidence.	Prediction Module	Done	Medium
5	Primary & Secondary Emotion Detection	Displays top predicted emotions.	Prediction Module	Done	Medium
6	Personalized Learning Guidance	Generates learning suggestions.	Guidance Module	Done	High
7	Emotion Logging	Stores prediction history in CSV.	Logging Module	Done	Medium
8	Analytics Dashboard	Displays emotion distribution/logs.	Dashboard Module	Done	High

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-have Features	6
Additional / Nice-to-have Features	2
Features Tested and Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface	2	Text input interface, Results display
2	Backend / Logic	3	Emotion prediction, NLP preprocessing, Learning guidance
3	Database / Storage	1	Emotion prediction logging (CSV)
4	API / Integration	1	Google Gemini API integration
5	Security / Authentication	1	Secure local processing of user data